

Connor Fuglestad

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GitHub: www.github.com/cfuglestad ; Portfolio: cfuglestad.github.io

Applied Data scientist and ML engineer with seven years of experience building production analytics systems in healthcare. I train and evaluate models (NER, knowledge distillation, sequence classification), design dimensional data models over billions of rows, and write the pipelines that move data between the two. MS Economics (econometrics focus) with graduate work in natural language technology. Most of my time goes to problems where statistical validity and engineering reliability both need to be right.

SKILLS

Machine Learning, NLP, and Statistical Methods: Knowledge Distillation, Fine-Tuning (QLoRA), NER, Token Classification, Transformer Models (BERT, PubMedBERT), Prompt Optimization (DSPy), RAG, Semantic Search, Time Series Forecasting (Prophet), Anomaly Detection, Econometrics, Regression Analysis, Causal Inference, Hypothesis Testing, A/B Testing, Model Evaluation (F1, Precision, Recall, MRR)

Frameworks and Libraries: PyTorch, HuggingFace Transformers, scikit-learn, LangGraph, LangChain, DSPy, ChromaDB, sentence-transformers, MLflow, PySpark, Apache Spark, pandas, NumPy

Data Engineering: Databricks (Notebooks, Jobs, SQL Warehouses, Unity Catalog), Delta Lake, ETL/ELT Pipeline Design, Incremental Loading, Dimensional Modeling (Star/Constellation Schema), Medallion Architecture, DAG Scheduling, Parallel Execution, SQL, Data Validation

Tools and Infrastructure: Python, Azure DevOps (CI/CD), Git, Power BI, REST API Integration, Serverless Compute, Unit Testing (pytest, sqlglot AST parsing)

EXPERIENCE

Vizient

Remote, USA

Analytics & Insights Lead

December 2025-Present

- Architected and deployed a 43-table, 592-column dimensional model (star/constellation schema) over 3.1 billion rows of hospital quality data from Vizient's national clinical database. The schema is the single analytic layer for executive Power BI dashboards, statistical process control alerting, and downstream ML feature extraction across 25+ acute care facilities.
- Replaced full-table TRUNCATE + INSERT with a date-windowed incremental load (DELETE + INSERT on a rolling 4-month discharge window) across 27 fact and dimension tables. Reduced the primary ETL task from 19.5 minutes to 8.4 minutes (57% faster) and cut total pipeline wall-clock time from 36.6 to 25.1 minutes with proportional compute cost reduction on a serverless SQL warehouse.
- Built a dependency-aware parallel execution framework in Python (ThreadPoolExecutor with topological sort from a YAML-declared DAG). The scheduler resolves inter-table dependencies into 5 execution waves running 8 concurrent threads, reducing the dimension load stage from 5.2 to 2.5 minutes and fact load from 19.5 to 8.4 minutes by saturating all available warehouse clusters.
- Implemented conditional task execution via content-addressable hashing (MD5 of the YAML metadata manifest stored as a Delta table property). On unchanged metadata, the 5.5-minute comment-application step is skipped entirely. Combined with incremental loads and parallelism, day-2+ pipeline runs complete in under 20 minutes versus 36.6 minutes at baseline.
- Designed SPC-based anomaly detection (Shewhart control charts with Western Electric decision rules) for 30+ clinical quality metrics at the facility-month grain. The system flags statistical outliers in mortality O:E ratios, 30-day readmission rates, excess days, and cost indices, writing results to a dedicated fact table that feeds automated Teams alerting and a seasonal overlay via Prophet for metrics with strong periodicity.
- Created a 452-million-row metric-encounter bridge table implementing 9 distinct metric construction patterns (mortality O:E, CMS condition-specific readmission, EDAC excess days, AHRQ PSI, continuous LOS/cost). Enables Power BI DirectQuery drill-through from aggregate quality scores to individual contributing encounters without data duplication.
- Authored 58 offline unit tests covering DDL/DML column parity validation, source-table contract enforcement via sqlglot abstract syntax tree parsing, job runtime least-privilege grant verification, and deployed schema-order contracts. Integrated with Azure DevOps CI for automated regression gating on every push.

Senior Analytics Engineer

April 2025-December 2025

- Migrated 20+ legacy Python ETL pipelines to Databricks (PySpark + Unity Catalog), adding schema enforcement, data lineage tracking, and modular transformation stages that reduced failure investigation time from hours to minutes.
- Built a reference-file-driven ingestion framework that auto-generates Spark SQL transformations from 40+ YAML configuration files. The system discovers new source columns, validates data contracts, and propagates schema changes without manual code edits. This architecture directly scaffolded the 43-table star schema built in the subsequent role.
- Prototyped AI-assisted analysis workflows (LLM-based summarization of clinical narratives, automated anomaly flagging on quality metrics) and integrated them into reporting pipelines as supplementary analytic signals for client health systems.

Analytics Engineer

November 2022-April 2025

- Designed the logical and physical data models (conformed dimensions, slowly changing type-2 history, star schema grain definitions) that became the foundation for all downstream Vizient analytics products. Modeled 40+ healthcare source tables into a governed reporting layer consumed by Power BI, statistical analyses, and quality reports across 25+ health systems.
- Developed a shared Python library (internal package) standardizing API-driven data acquisition, flat-file parsing, and transformation patterns across 20+ production products. Reduced onboarding time for new data sources from weeks to days.
- Automated data integration processes (REST API ingestion, flat-file parsing, cross-source reconciliation) using Python and SQL, eliminating approximately 70% of recurring manual effort while adding validation checks that caught upstream schema drift before it reached downstream reporting.
- Established the core data pipelines (ingestion, transformation, quality checks) that became production infrastructure for 4 Vizient analytics products serving both internal teams and external health system clients.

University of Wisconsin Health System
Analytics Consultant

Remote, USA
January 2022-November 2022

- Applied econometric methods (regression, time series forecasting, panel data models) to EHR-derived clinical and operational datasets using Python and SQL, producing demand forecasts and staffing optimization models adopted by department leadership for resource allocation decisions.
- Migrated UW Health's highest-traffic productivity dashboard to QlikSense, redesigning the underlying data model (normalized source to star schema) and reducing page load time from minutes to seconds for 200+ daily users.
- Automated a financial reporting workflow that previously required 8 hours of manual Excel manipulation. The Python + SQL replacement runs in 20 minutes, eliminated transcription errors, and freed analyst capacity for statistical modeling work.
- Built reproducible analytical workflows for clinical quality accreditation reporting (performance measures, compliance metrics, outcome tracking) using version-controlled SQL and Python.

Resolution Economics
Consultant

Chicago, IL and Remote, USA
April 2019-July 2020

- Constructed analytical datasets in SAS (millions of rows, joined payroll/census/wage survey sources) and applied regression analysis and hypothesis testing to quantify employment disparities for expert testimony in class action litigation.

PROJECTS

Github.com/cfuglestad

- Built a clinical NER knowledge distillation pipeline that trains a PubMedBERT token classifier (110M params) on GPT-4o-mini-generated annotations over synthetic clinical notes (Synthea). The student model targets 5 entity types (MEDICATION, CONDITION, PROCEDURE, LAB_VALUE, ANATOMY) using BIO sequence labeling and includes a separate assertion classifier (AFFIRMED, NEGATED, HYPOTHETICAL, FAMILY, HISTORICAL) with a NegEx-rules baseline. Evaluation compares F1, latency, and inference cost against the teacher model to quantify the accuracy-efficiency tradeoff of distillation for domain-specific NLP.
- Developed a resume tailoring system using LangGraph (cyclic state machine with conditional routing), DSPy signature-based prompt optimization, and QLoRA fine-tuning on a 4-bit quantized LLM. The pipeline implements extract, grade, and iterative tailor nodes with programmatic evaluation (precision, recall, F1 against gold-label keyword sets). Training data generated synthetically across 8 job-archetype templates (80 examples).
- Built a RAG system that answers clinical questions from published practice guidelines with explicit citations. LangGraph agent with confidence-based abstention, ChromaDB semantic retrieval (all-MiniLM-L6-v2), and a labeled evaluation framework measuring Precision@5, Recall@5, and MRR across difficulty-stratified test sets.

EDUCATION

University of Wisconsin – Madison
Master of Science, Economics

Madison, WI
September 2020-December 2021

Relevant Graduate Coursework: Econometrics I-III, Microeconomic Theory I & II, Macroeconomic Theory I, Machine Learning, Time Series Forecasting, Mathematics for Economists, Data Analytics for Economists

University of Wisconsin – Madison
Bachelor of Arts, Economics (Mathematical Emphasis) and German

Madison, WI
September 2015-December 2018

Relevant Undergraduate Coursework: Introductory Econometrics, Economic Forecasting, Statistics for Economics, Law and Economics, Economics of Health Care, Linear Algebra, Calculus I-III

University of Washington
Certificate in Natural Language Technology (In Progress)

Seattle, WA
July 2026-March 2027 (Expected)

Relevant Graduate Coursework: Shallow Processing Techniques for Natural Language Processing, Advanced Statistical Methods in Natural Language Processing, Computational Linguistic Theory